

IT3091 - Machine Learning

50 Multi-Select MCQs

Lab Sheets 06–08: Regression, SVM/SVR, Ensembles and Clustering

Suggested duration: 125 minutes
Number of questions: 50
Options per question: Five (A–E)
Question type: Select all correct answers

Instructions

- Every question has exactly **two or three** correct options.
- Select every correct option. A statement should be selected only if it is fully correct.
- Questions are mixed across Lab Sheets 06, 07 and 08; no answer key is included in this student paper.
- Assume standard Scikit-learn behaviour and the settings stated in the laboratory sheets.

01. Which statements correctly describe residuals and predictions in simple linear regression?

- ☐ A. A residual can be calculated as $y_i - \hat{y}_i$.
- ☐ B. A positive residual means that the model predicted above the actual value.
- ☐ C. Mean Squared Error is expressed in the same units as the target variable.
- ☐ D. Squaring residuals reduces the influence of observations with large errors.
- ☐ E. A zero residual means that the fitted value equals the observed value.

02. Why is feature scaling important when fitting SVM or SVR models?

- ☐ A. It prevents a large-range feature from dominating the geometry of the feature space.
- ☐ B. The scaler should be fitted on training data and then applied unchanged to test data.
- ☐ C. It converts every classification target into a continuous response.
- ☐ D. It guarantees that a linear kernel will perfectly separate the training classes.
- ☐ E. SVM and SVR decisions depend strongly on distances and relative feature magnitudes.

03. Which mechanisms create diversity among trees in a standard Random Forest?

- ☐ A. Each tree is trained using a bootstrap sample of the training observations.
- ☐ B. A random subset of features is considered when searching for a split.
- ☐ C. Each new tree is trained to correct the residuals of the preceding tree.
- ☐ D. Every tree is forced to use exactly the same structure and split variables.
- ☐ E. Each tree receives the full dataset sampled once without replacement.

04. Which operations belong to one normal K-Means iteration after centroids have been initialized?

- ☐ A. Use known class labels to correct incorrect assignments.
- ☐ B. Mark sparse observations with the permanent label -1 .
- ☐ C. Merge the two closest clusters using a linkage rule.
- ☐ D. Assign every observation to its nearest current centroid.
- ☐ E. Replace each centroid with the mean of its assigned observations.

05. Which practices correctly protect the test set during model development?

- ☐ A. Use cross-validation within the training data to select hyperparameters.
- ☐ B. Use the untouched test set for the final evaluation of the selected model.
- ☐ C. Try many hyperparameter settings on the test set and retain the best one.
- ☐ D. Standardize the entire dataset before creating the train-test split.
- ☐ E. Select the final model using training accuracy alone.

06. Which statements correctly describe gradient descent for the from-scratch linear-regression model?

- ☐ A. The coefficient update adds the learning rate multiplied by the gradient.
- ☐ B. The gradient indicates the direction in which the loss changes.
- ☐ C. The learning rate controls the size of each parameter update.
- ☐ D. Any positive learning rate is guaranteed to produce convergence.
- ☐ E. A correctly implemented training loss must increase at every iteration.

07. Which statements describe DBSCAN as used in Lab Sheet 08?

- ☐ A. It forms clusters by connecting observations in sufficiently dense regions.
- ☐ B. Scikit-learn uses the label -1 for observations classified as noise.
- ☐ C. It can recover non-spherical clusters such as interlocking moons.
- ☐ D. The number of clusters K must be specified before fitting.
- ☐ E. Every cluster is represented by a centroid equal to its mean.

08. For a fitted binary linear SVM with $f(x) = w^T x + b$, which statements are correct?

- ☐ A. Both margin boundaries satisfy $f(x) = 0$.
- ☐ B. Every training observation must become a support vector.
- ☐ C. The decision boundary satisfies $w^T x + b = 0$.
- ☐ D. The predicted class is $+1$ when $f(x) \geq 0$.
- ☐ E. The raw decision score is automatically a calibrated probability.

09. Which statements correctly describe agglomerative hierarchical clustering?

- ☐ A. It begins with all observations in one cluster and repeatedly splits them.
- ☐ B. It begins with each observation treated as its own cluster.
- ☐ C. It repeatedly merges the closest pair of current clusters.
- ☐ D. The final number of clusters must be chosen before constructing the hierarchy.
- ☐ E. It assigns points only by their distance to fixed mean centroids.

10. Which regression metrics retain the same measurement units as the target variable?

- ☐ A. Root Mean Squared Error
- ☐ B. Mean Squared Error
- ☐ C. Coefficient of determination R^2
- ☐ D. Mean Absolute Error after squaring its value
- ☐ E. Mean Absolute Error

11. Which statements characterize Gradient Boosting in the laboratory sheet?

- ☐ A. All trees are trained independently and can be fitted fully in parallel.
- ☐ B. Final predictions are obtained only by majority voting over bootstrap trees.
- ☐ C. Trees are added sequentially rather than independently.
- ☐ D. A new tree attempts to improve the prediction of the current ensemble.
- ☐ E. Small or shallow trees are commonly used as the individual learners.

12. Which statements about K-Means WCSS and the elbow approach are correct?

- ☐ A. K-Means attempts to maximize WCSS so that clusters remain distinct.
- ☐ B. WCSS sums squared distances from observations to their assigned centroids.
- ☐ C. The minimum achievable WCSS cannot increase when K is increased.
- ☐ D. The elbow suggests a trade-off between compactness and model complexity.
- ☐ E. Cluster labels should be compared numerically across runs to verify equality.

13. How does the regularization parameter C affect a soft-margin SVM?

- ☐ A. A very small C forces the classifier to fit every training point exactly.
- ☐ B. The value of C selects whether the kernel is linear or nonlinear.
- ☐ C. A smaller C accepts more margin violations in exchange for stronger regularization.
- ☐ D. A larger C penalizes training violations more heavily.
- ☐ E. A smaller C may be safer when a training point could be mislabelled.

14. Which statements describe the basic idea of ensemble learning?

- ☐ A. Diverse models can sometimes offset one another's uncorrelated errors.
- ☐ B. An ensemble is guaranteed to outperform every member on every dataset.
- ☐ C. An ensemble can be trained without any target in a supervised task.
- ☐ D. Classification outputs can be combined using voting.
- ☐ E. Regression outputs can be combined using averaging.

15. Which statements correctly describe the ε -tube in linear SVR?

- ☐ A. It converts the continuous target into two classification labels.
- ☐ B. Its boundaries are $f(x) + \varepsilon$ and $f(x) - \varepsilon$.
- ☐ C. An observation strictly inside the tube produces zero ε -insensitive loss.
- ☐ D. Observations on or outside the tube can act as support vectors.
- ☐ E. The loss equals the absolute error without subtracting ε .

16. What may happen when `n_estimators` is increased in a Random Forest while other settings stay fixed?

- ☐ A. Trees begin to depend sequentially on the errors of earlier trees.
- ☐ B. Training accuracy must decrease after every additional tree.
- ☐ C. Test accuracy must increase strictly for every additional tree.
- ☐ D. The ensemble's predictions often become more stable.
- ☐ E. Performance improvements commonly level off after enough trees are added.

17. Why is StandardScaler useful before distance-based clustering?

- ☐ A. It centers and rescales features to comparable numerical scales.
- ☐ B. It reduces the chance that a large-range feature dominates Euclidean distance.
- ☐ C. It preserves every original pairwise Euclidean distance exactly.
- ☐ D. It creates the missing target labels required by unsupervised learning.
- ☐ E. It scales only the response variable while leaving predictors unchanged.

18. For binary SVM classification, which statements about hinge loss are correct?

- ☐ A. Every correctly classified observation has zero hinge loss.
- ☐ B. Hinge loss becomes negative for a point far beyond the margin.
- ☐ C. The individual loss is $\max(0, 1 - y_i f(x_i))$.
- ☐ D. The loss is zero when $y_i f(x_i) \geq 1$.
- ☐ E. Only incorrectly classified observations can have positive hinge loss.

19. Which actions are appropriate before assigning business names to discovered customer clusters?

- ☐ A. Assess whether similar groupings appear under reasonable changes in settings.
- ☐ B. Treat each numeric cluster ID as an already validated customer segment.
- ☐ C. Choose the solution with the largest WCSS because it separates groups most.
- ☐ D. Assume the groups are true classes because the algorithm converged.
- ☐ E. Inspect the feature profiles and practical meaning of every cluster.

20. The Gradient Boosting grid has two values each for n_estimators, learning_rate, and max_depth, with 5-fold cross-validation. Which statements are correct?

- ☐ A. The test set should be examined after every candidate combination.
- ☐ B. The best setting should be the one with the highest training accuracy.
- ☐ C. Only one fitted model is needed because all parameters are evaluated together.
- ☐ D. There are $2 \times 2 \times 2 = 8$ hyperparameter combinations.
- ☐ E. Each combination is evaluated across five validation folds.

21. Which statements match the from-scratch simple linear-regression model in Lab Sheet 06?

- ☐ A. Its primary output is a class obtained from the sign of $\hat{y}$.
- ☐ B. The intercept β_0 is the predicted value when $x = 0$.
- ☐ C. The slope β_1 is the predicted change for a one-unit increase in x .
- ☐ D. Its prediction is $\hat{y} = \beta_0 + \beta_1 x$.
- ☐ E. The slope is defined as the residual of the first observation.

22. Which conclusions about max_depth are appropriate for tree-based models?

- ☐ A. It sets the number of trees included in the ensemble.
- ☐ B. It controls the fraction of features considered at each split.
- ☐ C. It is selected by repeatedly maximizing accuracy on the test set.
- ☐ D. Increasing it usually allows an individual tree to fit training data more closely.
- ☐ E. Excessively deep trees can increase the risk of overfitting.

23. Which definitions of hierarchical-clustering linkage are correct?

- ☐ A. Average linkage uses only the distance between the two cluster centroids.
- ☐ B. Single linkage uses the farthest pair of observations across two clusters.
- ☐ C. Single linkage uses the closest cross-cluster pair of observations.
- ☐ D. Complete linkage uses the farthest cross-cluster pair of observations.
- ☐ E. Complete linkage always merges the clusters containing the fewest points.

24. If ϵ is increased in the SVR experiment while C remains fixed, which outcomes are reasonable?

- ☐ A. The number of support vectors often decreases because more points lie inside the tube.
- ☐ B. The fitted line and all predictions must remain exactly unchanged.
- ☐ C. The width of the no-penalty tube becomes smaller.
- ☐ D. The upper and lower tube boundaries move farther apart.
- ☐ E. More observations may receive zero ϵ -insensitive loss.

25. What does `max_features` control in a Random Forest?

- ☐ A. The number of observations sampled into every bootstrap dataset.
- ☐ B. The contribution of each new tree to the current ensemble.
- ☐ C. The number or fraction of predictors considered at a split.
- ☐ D. The amount of correlation among trees can change when this setting changes.
- ☐ E. Choosing the largest value is guaranteed to maximize test accuracy.

26. Which statements about R^2 are correct?

- ☐ A. A poor model evaluated on new data can have a negative R^2 .
- ☐ B. RMSE, rather than R^2 , is always restricted to the interval $[0, 1]$.
- ☐ C. MAE penalizes a large residual more strongly than squaring that residual.
- ☐ D. R^2 compares the residual sum of squares with total variation around the mean.
- ☐ E. $R^2 = 1$ indicates perfect predictions for the evaluated observations.

27. Which statements explain the role of a separate test set?

- ☐ A. Its observations should be included when fitting the final model coefficients.
- ☐ B. It approximates performance on observations not used during development.
- ☐ C. Repeatedly selecting models using test results turns the test set into tuning data.
- ☐ D. Competing final models should be compared on the same test observations.
- ☐ E. It should replace cross-validation for every hyperparameter decision.

28. Which statements address initialization in K-Means?

- ☐ A. A fixed random seed helps reproduce the same initialization.
- ☐ B. Different initial centroids can lead to different local solutions.
- ☐ C. Using multiple initializations can reduce dependence on one unlucky start.
- ☐ D. Every updated centroid must remain one of the original observations.
- ☐ E. Initialization cannot matter once more than one cluster is requested.

29. What can reasonably be inferred from Random Forest `feature_importances_`?

- ☐ A. It proves that changing an important feature will causally change the outcome.
- ☐ B. It gives the positive or negative direction of each feature's class effect.
- ☐ C. It directly measures final test accuracy for every individual feature.
- ☐ D. It summarizes how much features contributed to splitting within the fitted forest.
- ☐ E. Changing importance rankings across seeds may indicate correlated substitute predictors.

30. For the Gradient Boosting update $F_m(x) = F_{m-1}(x) + \eta h_m(x)$, let $F_{m-1}(x) = 60$, $h_m(x) = 10$, and $\eta = 0.1$. Which statements are correct?

- ☐ A. The updated prediction is 61.
- ☐ B. The updated prediction is 70.
- ☐ C. The updated prediction is 60.1.
- ☐ D. The new tree subtracts a correction of 1.
- ☐ E. The correction contributed by the new tree is 1.

31. Which steps are suitable when verifying the from-scratch regression model with Scikit-learn?

- ☐ A. Compare the learned intercept and coefficient with the from-scratch estimates.
- ☐ B. Compare both models' prediction for a student who studies 4.5 hours.
- ☐ C. Replace `LinearRegression` with `SVC` because both fit straight lines.
- ☐ D. Use only R^2 and ignore whether predictions and parameters agree.
- ☐ E. Expect the two implementations to produce intentionally opposite slopes.

32. How do the practical DBSCAN parameters affect the clustering result?

- ☐ A. `eps` directly specifies the exact number of clusters to create.
- ☐ B. DBSCAN is restricted to spherical clusters for every parameter setting.
- ☐ C. An excessively small `eps` can fragment clusters and label many points as noise.
- ☐ D. An excessively large `eps` can merge distinct dense regions.
- ☐ E. `min_samples` contributes to the requirement for a region to be considered dense.

33. Which statements correctly describe support vectors in a fitted linear SVM?

- ☐ A. They are always all training observations, regardless of the value of C .
- ☐ B. They are the observations on or within the margin, or violating it, that constrain the solution.
- ☐ C. Non-support observations lying safely beyond the margin do not determine the final boundary directly.
- ☐ D. They are selected because they are farthest from the decision boundary.
- ☐ E. They must have the smallest value of the first input feature.

34. Which statements about the Gradient Boosting learning rate are correct?

- ☐ A. A smaller learning rate may require more trees to achieve a similar fit.
- ☐ B. An excessively large learning rate can make corrections too aggressive and harm generalization.
- ☐ C. A learning rate of zero gives every new tree its maximum possible contribution.
- ☐ D. The setting is required only for Random Forest, not Gradient Boosting.
- ☐ E. It scales the contribution made by each newly added tree.

35. Why should K-Means label numbers not be compared directly across separate runs?

- ☐ A. Cluster 0 is mathematically required to be the most important cluster.
- ☐ B. The same grouping may be represented by a permutation of cluster IDs.
- ☐ C. Membership patterns or centroid locations are more meaningful than the numeric IDs.
- ☐ D. The cluster IDs are known target labels supplied during supervised training.
- ☐ E. Permuting label numbers necessarily changes WCSS.

36. Which statements explain why a Random Forest can generalize better than one unpruned Decision Tree?

- ☐ A. Every forest tree is required to be a one-level decision stump.
- ☐ B. Forest trees are added sequentially to correct the previous tree's residuals.
- ☐ C. A single deep tree can have high variance and be sensitive to small data changes.
- ☐ D. Combining diverse trees can reduce instability through aggregation.
- ☐ E. Forcing identical samples and split choices would weaken the benefit of averaging.

37. Which statements correctly describe ϵ -insensitive loss in SVR?

- ☐ A. It is the squared residual for every training observation.
- ☐ B. It can be written as $\max(0, |y - f(x)| - \epsilon)$.
- ☐ C. It equals zero when the absolute error is no greater than ϵ .
- ☐ D. It increases only for the portion of absolute error extending beyond the tube.
- ☐ E. It becomes negative when a prediction lies near the centre of the tube.

38. Which statements help interpret an agglomerative-clustering dendrogram?

- ☐ A. The vertical fusion height represents the dissimilarity at which clusters merge.
- ☐ B. A horizontal cut determines a partition and therefore a number of clusters.
- ☐ C. A merge at a larger height joins clusters that were less similar.
- ☐ D. The final number of clusters must be supplied before the dendrogram is created.
- ☐ E. Changing linkage cannot change the merge order or dendrogram structure.

39. Which statements distinguish supervised learning from clustering in the lab?

- ☐ A. After clustering, feature profiles should be examined before assigning real-world meanings.
- ☐ B. The hidden labels returned by `make_blobs` should be used to train K-Means.
- ☐ C. A discovered cluster is automatically a verified true class.
- ☐ D. Supervised learning uses an available target to learn an outcome prediction.
- ☐ E. Clustering seeks structure when no target labels are available for training.

40. Which statements correctly compare Random Forest construction with Gradient Boosting construction?

- ☐ A. Random Forest trees use bootstrap samples of observations.
- ☐ B. Random Forest introduces random feature selection during splitting.
- ☐ C. Random Forest trees are trained sequentially on the preceding forest's residuals.
- ☐ D. A learning-rate parameter scales every Random Forest tree's vote.
- ☐ E. Random Forest trees are designed to be trainable independently rather than as a correction sequence.

41. Which statements correctly describe Mean Squared Error?

- ☐ A. It is the average of the squared residuals.
- ☐ B. A large residual has disproportionately greater influence than under MAE.
- ☐ C. It can become negative when most predictions exceed the observations.
- ☐ D. It is expressed in exactly the same units as the target.
- ☐ E. It allows positive and negative residuals to cancel before squaring.

42. Which statements concern the centroid-update and stopping stages of from-scratch K-Means?

- ☐ A. A cluster centroid must be the coordinate-wise median of assigned observations.
- ☐ B. Known target labels are required to determine whether convergence occurred.
- ☐ C. The mean minimizes the within-cluster sum of squared Euclidean distances.
- ☐ D. An implementation should handle the possibility that a cluster receives no observations.
- ☐ E. Training can stop when centroids change negligibly or the iteration limit is reached.

43. Which Gradient Boosting hyperparameter interpretations match Lab Sheet 07?

- ☐ A. `n_estimators` controls the number of sequential trees.
- ☐ B. `max_depth` controls the complexity of each individual tree.
- ☐ C. `learning_rate` controls the contribution of each new tree.
- ☐ D. `max_features` must always equal the total number of observations.
- ☐ E. Increasing all three settings is guaranteed to improve test performance.

44. For $A = (2, 3)$ and $B = (8, 11)$, which statements are correct?

- ☐ A. A feature measured on a much larger numerical scale can dominate Euclidean distance.
- ☐ B. Their Euclidean distance is $\sqrt{8^2 + 11^2}$.
- ☐ C. Their Euclidean distance is 14.
- ☐ D. Standardization would leave every pairwise distance numerically identical.
- ☐ E. Their Euclidean distance is $\sqrt{(8-2)^2 + (11-3)^2} = 10$.

45. Which statements describe the margin of a fitted canonical linear SVM?

- ☐ A. The two margin boundaries are $w^T x + b = +1$ and $w^T x + b = -1$.
- ☐ B. The distance between the two margin boundaries is $2/\|w\|$.
- ☐ C. A large margin is preferred because it can improve robustness and generalization.
- ☐ D. The decision boundary is the union of the two equations $w^T x + b = \pm 1$.
- ☐ E. The margin width is numerically equal to C for every fitted dataset.

46. Which statements correctly compare Random Forest and Gradient Boosting?

- ☐ A. Gradient Boosting commonly uses a learning rate to shrink each new correction.
- ☐ B. Both methods form predictions by combining multiple trees.
- ☐ C. Random Forest trees must learn directly from the predictions of earlier trees.
- ☐ D. Gradient Boosting is simply an average of independent bootstrap trees.
- ☐ E. Random Forest does not normally require a boosting-style learning rate.

47. When clustering the interlocking-moons dataset, which comparisons are appropriate?

- ☐ **A.** K-Means explicitly labels isolated observations as noise using -1 .
- ☐ **B.** DBSCAN can follow dense curved regions without representing them by one centroid each.
- ☐ **C.** DBSCAN can explicitly distinguish noise observations from cluster members.
- ☐ **D.** Both methods require the number of clusters K as a mandatory input.
- ☐ **E.** K-Means is guaranteed to recover the two moon shapes because $K = 2$.

48. Which actions or observations are consistent with controlling overfitting in Lab Sheet 07?

- ☐ **A.** Limiting tree depth can reduce individual-tree complexity.
- ☐ **B.** The model with the highest training accuracy should always be selected.
- ☐ **C.** The test set should be repeatedly consulted while tuning depth and learning rate.
- ☐ **D.** Near-perfect training accuracy with noticeably lower test accuracy is evidence of possible overfitting.
- ☐ **E.** Cross-validation provides a better tuning criterion than training accuracy alone.

49. After fitting `SVC(kernel='linear')`, which statements about manual decision scores are correct?

- ☐ **A.** A linear SVC never exposes fitted coefficients or an intercept.
- ☐ **B.** The intercept contains one separate value for every training observation.
- ☐ **C.** For the binary linear model, $\mathbf{X} @ \mathbf{w} + \mathbf{b}$ should match `decision_function(X)`.
- ☐ **D.** The sign of the score determines the predicted -1 or $+1$ class in the lab example.
- ☐ **E.** Every decision score is already a probability between zero and one.

50. Which criteria are appropriate when judging whether a clustering solution is valuable?

- ☐ **A.** Interpretability: the clusters can be explained using meaningful feature patterns.
- ☐ **B.** Usefulness: the grouping reveals structure relevant to the practical problem.
- ☐ **C.** The solution with the greatest number of clusters should always be preferred.
- ☐ **D.** Any converged clustering must be accepted as the true population grouping.
- ☐ **E.** Stability: similar structure appears under reasonable repetitions or setting changes.